CS5234 Quiz 2 · Cheat Sheet

L6–L11 · k-means not examined · log = ln

0 · Recognise the problem → tool

- **Streaming upper bound**: keep a sketch / sample / summary; unbiased estimator → variance → Chebyshev (mean trick) → median trick
- **Streaming lower bound**: one-way communication; reduce from Index / Gap-Hamming / Disjointness; encode Bob's index as a key edge / element
- **Deletions allowed (turnstile)**: use a linear sketch Rx
- **Unknown scale**: convert to a testing problem with threshold T, run $T = (1+\varepsilon)^j$ in parallel
- **Approximation algorithm**: greedy + triangle inequality; pigeonhole on optimal clusters
- **How much independence?** Chebyshev only → pairwise; 4th moment (AMS) → 4-wise; conditioning on k heavy items → $(k+2)$-wise

1 · Toolbox recap

- Markov $\Pr(X \geq t) \leq \mathbb{E}X/t$; Chebyshev $\Pr(|X - \mathbb{E}X| \geq t) \leq \mathrm{Var}X/t^2$
- Mean of k i.i.d.: Var/k; need $k = \frac{3\mathrm{Var}X}{\varepsilon^2(\mathbb{E}X)^2}$ for 2/3
- Chernoff (indep., $[0,1]$): $\Pr(|X-\mu| \geq \delta\mu) \leq 2e^{-\delta^2\mu/3}$
- Median trick: success $> \frac{1}{2}$ → $O(\log \frac{1}{\delta})$ copies; union bound over m items with $\delta = 1/m^2$
- $\{0,1\}$ var.: $\mathrm{Var} \leq \mathbb{E}$; pairwise indep. ⇒ variances add
- $(1-\frac{1}{T})^T \approx \frac{1}{e}$, $(1-\frac{1}{T})^{2T} \leq \frac{1}{e^2}$; $\frac{d^k}{k!} \leq (\frac{ed}{k})^k$

2 · L6 Clustering

Setup metric d: $d(p,p)=0$, symmetric, triangle inequality. Choose k centres; $C(p)$ = nearest centre. k-center: $\min \max_i d(p_i, C(p_i))$; k-median: $\min \sum_i d(p_i, C(p_i))$.

Hardness 1-2 metric (distances in $\{1,2\}$, triangle inequality automatic). Dominating set of size $k \Leftrightarrow k$-center OPT ≤ 1 (edge → 1, non-edge → 2) ⇒ ratio < 2 is NP-hard.

Farthest-first greedy = 2-approx S_i^* optimal clusters, $D(S_i^*) \leq OPT$. Point $p \in S_i^*$: (a) greedy picked $c \in S_i^*$ ⇒ $d(p,c) \leq d(p,C_i^*) + d(C_i^*, c) \leq 2OPT$; (b) none picked in S_i^* ⇒ k centres in $k-1$ clusters ⇒ two centres C_{ℓ_1}, C_{ℓ_2} ($\ell_1 < \ell_2$) share a cluster, $d \leq 2OPT$; when C_{ℓ_2} was chosen it was the farthest point ⇒ every point within $2OPT$ then (and later). Query: nk distance queries.

Streaming k-center Test(T): first point is a centre; new p farther than $2T$ from all centres → new centre, or "No" if already k. Output centres ⇒ cost $\leq 2T$. "No" ⇒ $k+1$ points pairwise $> 2T$ ⇒ each radius-T ball covers ≤ 1 ⇒ $OPT > T$. Run $T = (1+\varepsilon)^j$: smallest non-"No" T^* has $OPT/2 \leq T^* \leq (1+\varepsilon)OPT$ ⇒ $(2+2\varepsilon)$-approx; space $O(k\frac{\log \Delta}{\varepsilon})$ points.

k-median coreset ($< 1 + \frac{2}{e}$ NP-hard; best $2+\varepsilon$.) Blocks of $\sqrt{nk}$ points, α-approx each → k weighted centres into S ($O(\sqrt{nk})$ points) → α-approx on S.

Lemma $OPT(P) \leq 2OPT^*(P)$ (move each free centre to nearest input point).

$\sum d(p_i, C_i) \leq \sum d(p_i, s_i) + \sum d(s_i, C_i)$;
$\sum d(p_i, s_i) \leq \alpha \sum_j OPT(P_j) \leq 2\alpha OPT$;
$\sum d(s_i, C_i) \leq 2\alpha \sum d(s_i, C_i^*) \leq 2\alpha(2\alpha+1)OPT$
⇒ $(4\alpha^2+4\alpha)$. L levels: $(4\alpha+4)^L$, $O(Lkn^{1/L})$ points.

MEB (1-center) Some input point on the circle in every half-plane through the centre. Greedy: add farthest point, recompute MEB; $\lambda_i = r_i/D$, $\lambda_{i+1} \geq \frac{1+\lambda_i^2}{2}$ (max of $1-a$ and $\sqrt{\lambda_i^2+a^2}$); $x_i = 1-\lambda_i$, $\frac{1}{x_{i+1}} \geq \frac{1}{x_i} + \frac{1}{2}$ ⇒ $t > \frac{10}{\varepsilon}$ gives $(1+\varepsilon)$; coreset of $O(1/\varepsilon)$ points.

3 · L7 Distinct elements

Testing $h : [m] \to [T]$; $\Pr[\text{no element hashes to } 1] = (1-\frac{1}{T})^{DE}$: $\approx \frac{1}{e}$ if $DE \leq T$, $\leq \frac{1}{e^2}$ if $DE \geq 2T$. $k = O(1)$ copies, threshold $\frac{1}{2}(\frac{1}{e} + \frac{1}{e^2})$, Chernoff. Guess $T = (1+\varepsilon)^j$ ⇒ $(2+2\varepsilon)$. Fully random h costs $m \log T$ bits.

KMV $h : [m] \to [m]$, $t = \frac{100}{\varepsilon^2}$, X = t-th smallest hash, output $\widehat{DE} = \frac{tm}{X}$.

Under-estimate ⇔ fewer than t hashes below $\frac{tm}{(1-\varepsilon)DE}$; $Y = \sum Y_i$, $\mathbb{E}Y = \frac{t}{1-\varepsilon}$, $\mathrm{Var}Y \leq \mathbb{E}Y$ (pairwise) ⇒ Chebyshev $\leq \frac{1}{10}$; over-estimate same; union ⇒ 0.8; median ⇒ $1-\delta$. Space $O(\varepsilon^{-2} \log m \log \frac{1}{\delta})$. **!** fails under deletions

k-wise independent hash

$\Pr[h(x_1)=y_1, \ldots, h(x_k)=y_k] = m^{-k}$ for distinct x's. $h(x) = \sum_{j<k} c_j x^j$ over $\mathbb{F}_m$ (m prime). Proof: two polynomials agreeing on k points differ by a degree-$\leq k-1$ poly with k roots ⇒ equal; m^k polys ↔ m^k output tuples (bijection). Space $O(k \log m)$.

4 · L8 Communication complexity

Streaming → one-way CC Alice runs the algorithm on the first half, sends memory (S bits); Bob continues. CC lower bound C ⇒ space $\geq C$.

Equality det. $\geq n$: $\leq n-1$ bits ⇒ $\leq 2^{n-1}$ messages $< 2^n$ inputs ⇒ $x_1 \neq x_2$ same message; Bob with $y = x_1$ fails on one. Rand.: send $r \cdot x$ mod 2; $x \neq y$ caught w.p. $\frac{1}{2}$; $\log \frac{1}{\delta}$ repeats; needs shared r.

Newman public → private with $+O(\log n + \log \frac{1}{\delta})$ bits, error $+\delta$: fix $t = 100n/\delta^2$ strings; Chernoff per input ($< 2^{-10n}$) + union over 2^{2n} inputs; Alice sends index i.

Hard problems Index (x / $i \to x_i$): $\Omega(n)$. Gap-Hamming (Hamming distance $\pm\sqrt{n}$): $\Omega(n)$.

Reduction A hard (C), extra message $|M|$ ⇒ B needs $C - |M|$.

Set intersection $\pm\frac{\sqrt{n}}{2}$: Ham $= |X|+|Y|-2|X \cap Y|$, send $|X| \Rightarrow \Omega(n)$.

Disjointness: $X = \{j : x_j = 1\}$, $Y = \{i\}$ ⇒ $\Omega(n)$.

Graph streams Connectivity $\Omega(n)$: u_A-v_j iff $x_j = 1$ (Alice); u_B-v_j $\forall j \neq i$, u_A-u_B (Bob) ⇒ connected ⇔ $x_i = 1$. Perfect matching $\Omega(n^2)$: Index on n^2 bits as bipartite u_i-v_j; Bob attaches private dummies to all $u_{i\neq i^*}$, $v_{j\neq j^*}$ ⇒ PM ⇔ edge $u_{i^*}v_{j^*}$. MIS $\Omega(n^2)$ (two copies + special vertices).

5 · L9 Frequency estimation

Moments $F_p = \sum f_i^p$; F_0=distinct, F_1=length. Collision sampling needs $\Theta(\sqrt{n}/\varepsilon^2)$ — too many.

AMS (F_2) $p_i \in \{\pm 1\}$, $Z = \sum p_i f_i$ (update $Z \mathrel{+}= p_x \Delta$), output Z^2.

$\mathbb{E}Z^2 = \sum f_i^2$ (cross terms vanish); $\mathbb{E}Z^4 = \sum f_i^4 + 3\sum_{i\neq j} f_i^2 f_j^2 \leq 3F_2^2$ ⇒ $\mathrm{Var}Z^2 \leq 2F_2^2$. Average $30/\varepsilon^2$ copies ⇒ Chebyshev $\leq \frac{1}{10}$. 4-wise signs. Space $O(\varepsilon^{-2}(\log m + \log n))$.

Models $\Delta = 1$ frequency; $\Delta \geq 0$ cash register; any turnstile; $x_i \geq 0$ always strict turnstile.

Linear sketch keep $y = Rx$; update $y \mathrel{+}= \Delta\, Re_i$ ⇒ deletions free. AMS ✓, KMV ✗. F_p ($p > 2$): $\tilde{\Theta}(m^{1-2/p})$.

Misra-Gries $\leq k$ counters; present → $+1$; room → add; full → all -1, drop zeros. $\leq n/k$ decrement steps ⇒ $x_i - \frac{n}{k} \leq c_i \leq x_i$. $k = 1/\varepsilon$: error εn, deterministic, $O(\varepsilon^{-1}(\log m + \log n))$, insert-only, not linear. (Majority: one pair (i, c).)

Count-Min t rows $\times k$; $C_\ell[h_\ell(i)] \mathrel{+}= \Delta$; $\tilde{x}_i = \min_\ell$. Over-count mean $\leq \|x\|_1/k$; $k = 2/\varepsilon$ + Markov ⇒ row bad w.p. $< \frac{1}{2}$; $t = 2\log_2 m$ ⇒ $1/m^2$; union ⇒ $x_i \leq \tilde{x}_i \leq x_i + \varepsilon\|x\|_1$ $\forall i$. Strict turnstile only; pairwise hash.

Count Sketch also $g_\ell : [m] \to \{\pm 1\}$; $C_\ell[h_\ell(i)] \mathrel{+}= g_\ell(i)\Delta$; $Y_{\ell,i} = g_\ell(i) C_\ell[h_\ell(i)]$: $\mathbb{E} = x_i$, $\mathrm{Var} \leq \|x\|_2^2/k$; $k = 10/\varepsilon^2$ ⇒ 0.9; median of $t = 100 \log m$ ⇒ $|\tilde{x}_i - x_i| \leq \varepsilon\|x\|_2$ $\forall i$. Any turnstile.

ℓ_1 heavy hitters → Count-Min; ℓ_2 → Count Sketch.

Sparse recovery $\|x\|_0 \leq k$: lower bound $\Omega(k(\log m + \log n))$. 1-sparse: value $= \sum \Delta$; position bit b = [sum over indices with bit b set > 0] (binary search). General: random 0/1 r separates $x \neq y$ w.p. $\geq \frac{1}{2}$; $O(k(\log m + \log n))$ fixed r's (exp. time).

6 · L10 Compression

Approx. sparse recovery $\mathrm{err}_2^k(x) = \min_{\|z\|_0 \leq k} \|z - x\|_2$ (keep top k). Count Sketch with $S = \frac{10k}{\varepsilon^2}$, $t = 100 \log m$; keep top-k of x'.

Step 1: no collision with the k large items w.p. $\geq 1 - \frac{\varepsilon^2}{10}$; then $\mathrm{Var} \leq \mathrm{err}^2/S$ ⇒ $|x'_i - x_i| \leq \frac{\varepsilon}{\sqrt{k}}\mathrm{err}$ ($(k+2)$-wise).

Step 2: I = chosen, J = true top-k; $j \in J \setminus I$ displaced by $i \in I \setminus J$: $|x_j| \leq |x_i| + \frac{2\varepsilon}{\sqrt{k}}\mathrm{err}$ ⇒ total $\leq (1+9\varepsilon)\mathrm{err}^2$ ⇒ $(1+5\varepsilon)\mathrm{err}$. Space $O(\varepsilon^{-2}k \log m(\log n + \log m))$.

JL $\Pi_{ij} \sim \mathcal{N}(0,1)$, $k = \frac{\alpha}{\varepsilon^2} \log \frac{1}{\delta}$: $\|\frac{1}{\sqrt{k}}\Pi x\|_2 = (1 \pm \varepsilon)\|x\|_2$ w.p. $1-\delta$ (independent of m). $y_i \sim \mathcal{N}(0, \|x\|_2^2)$; $\Pr[(1-\varepsilon)^2 k \leq \sum z_i^2 \leq (1+\varepsilon)^2 k] \geq 1 - 2e^{-c\varepsilon^2 k}$. n points: $\delta = \frac{1}{10n^2}$ + union ⇒ $k = O(\frac{\log n}{\varepsilon^2})$.

Regression $L(x) = \|Ax - b\|_2 = \|My\|_2$, $M = [A, b]$, $y = (x, -1)$. If $\|\Phi My\| = (1 \pm \frac{\varepsilon}{10})\|My\|$ $\forall y$: $(1-\frac{\varepsilon}{10})L(\tilde{x}) \leq \tilde{L}(\tilde{x}) \leq \tilde{L}(x^*) \leq (1+\frac{\varepsilon}{10})L(x^*)$ ⇒ $(1+\varepsilon)$.

Subspace embedding d-dim E, $k = O(\frac{d+\log(1/\delta)}{\varepsilon^2})$ ⇒ all $x \in E$ preserved. Unit vectors only; $\frac{1}{2}$-net $|N| \leq 5^d$ (disjoint radius-$\frac{\gamma}{2}$ balls inside radius $1+\frac{\gamma}{2}$); JL on N and differences ($2^{O(d)}$ events) ⇒ inner products; $x = \sum z_i$, $\|z_i\| \leq 2^{1-i}$, $\sum \|z_i\| \leq 2$ ⇒ error $O(\varepsilon)$. Sketch size $\tilde{O}(d^2/\varepsilon^2)$.

7 · L11 Graph algorithms

AGM x_v over $\binom{n}{2}$ pairs: $-w$ if $v = u_1$, $+w$ if $v = u_2$ ($u_1 < u_2$). $x_S = \sum_{v\in S} x_v$ is non-zero exactly on edges leaving S ⇒ ℓ_0-sample $Rx_S = \sum Rx_v$. $\log n$ rounds (components halve), fresh sketch per round (independence), ℓ_0-sampler $O(\log^2 n \log \frac{1}{\delta})$, $\delta = n^{-2}$ ⇒ $O(n \log^4 n)$.

MST stream add e; if cycle, delete heaviest edge on it. Cycle property: max edge of a cycle is in no MST (some MST avoids it). $O(n)$ edges.

MST query (weight only) weights in $[W]$, G_i = edges of weight $\le i$: **MST** $= n - W + \sum_{i=1}^{W-1} \#CC(G_i)$ ($W = 2$: $n - 2 + \#CC(G_1)$). $\#CC$ via random truncation r (Pr$[r \ge i] = \frac{1}{i}$), $x = \mathbf{1}[|CC(v)| \le r]$: bias $\le \frac{\varepsilon}{W}$, Var $\le \frac{\#CC}{n}$; sum the $W-1$ estimators, total variance $\le 2\varepsilon^2 \text{MST}^2$ ⇒ Chebyshev; $O(\varepsilon^{-2} nW \log \frac{W}{\varepsilon})$ pair queries. MST $\ge n-1$ ⇒ additive εn is relative.

MIS (Luby) random $L(v)$; add all local minima; delete them + neighbours. $X_{wu} = 1$ if $L(w)$ is min over $N(w) \cup N(u)$: $\mathbb{E} \ge \frac{1}{d_w + d_u}$, removes d_u edges; each edge counted ≤ 2 times ⇒ $\mathbb{E}[\text{removed}] \ge \frac{1}{2}\sum(\frac{d_u}{d_w+d_u} + \frac{d_w}{d_w+d_u}) = \frac{|E|}{2}$ ⇒ $10 \log n$ rounds + Markov. Multi-pass: 2 passes/round, $O(\log n)$ passes, $O(n \log n)$ bits. One-pass needs $\Omega(n^2)$.

Matching maximal $\ge \frac{1}{2}$ maximum. Stream greedy: $O(n)$ edges, 2-approx. Query: random $L(e)$ (sampled lazily); $Q(v)$ checks edges of v in increasing L; $Q(v, e{=}uv)$ = no earlier edge e' of u with $Q(u, e')$; sample $t = \frac{100}{\varepsilon^2}$ vertices, output $\frac{Xn}{2t}$. Recursion follows L-decreasing paths: Pr $\le \frac{d^k}{k!}$, depth $O(d/\varepsilon)$ ⇒ $d^{O(d/\varepsilon)}$ queries.

8 · Pitfalls

- Hash functions are fixed: repeated elements hit the same bucket (that is why hashing counts *distinct* elements).
- State the independence your analysis needs (pairwise / 4-wise / $(k{+}2)$-wise) and count its space.
- Count-Min needs $x_i \ge 0$; Misra-Gries is insert-only; KMV breaks under deletions.
- Lower-bound reductions: check both directions (answer 1 ⇔ $x_i = 1$) and that Alice's part depends only on x, Bob's only on i.
- Union bound over "all i" or "all inputs": per-item failure must be $\ll 1$/(number of items).

9 · Sketch comparison

Alg.	Model	Guarantee	Space
KMV	insert	$(1 \pm \varepsilon)DE$	$\varepsilon^{-2} \log m$
AMS	turnstile	$(1 \pm \varepsilon)F_2$	$\varepsilon^{-2} \log(mn)$
M-G	insert	$-\varepsilon n$, det.	$\varepsilon^{-1} \log(mn)$
CM	strict	$+\varepsilon\|x\|_1$	$\varepsilon^{-1} \log m \log(mn)$
CS	turnstile	$\pm\varepsilon\|x\|_2$	$\varepsilon^{-2} \log m \log(mn)$
JL	linear	$(1 \pm \varepsilon)\|x\|_2$	$\varepsilon^{-2} \log \frac{1}{\delta}$ rows
AGM	turnstile	spanning forest	$n \log^4 n$

KMV/AMS: $\times \log \frac{1}{\delta}$ for $1 - \delta$. M-G = Misra-Gries, CM = Count-Min, CS = Count Sketch.

10 · Answer templates

(A) Estimator analysis

1. Define indicator / signed variables; compute $\mathbb{E}$ (show unbiased or bound the bias).
2. Compute Var (expand square; cross terms vanish by pairwise / 4-wise independence).
3. Average k copies; Chebyshev ⇒ constant success.
4. Median of $O(\log \frac{1}{\delta})$ ⇒ $1 - \delta$; union bound if needed for all items.
5. Space = copies × (counter bits + hash seed bits).

(B) Lower bound by reduction

1. Pick source: Index (one bit), Gap-Hamming (approximate count), Disjointness.
2. Alice turns x into stream part 1; Bob turns i into part 2.
3. Alice runs alg., sends memory; Bob finishes and reads answer.
4. Prove: output YES ⇔ $x_i = 1$ (both directions).
5. Conclude: space $\ge$ CC $= \Omega(\cdot)$; mention randomness ⇒ randomized CC bound.

(C) Approximation ratio Fix an optimal solution; split into cases (pigeonhole: did the algorithm hit this optimal cluster?); chain triangle inequalities; sum over points / clusters.

11 · Quick numbers

- $\frac{1}{e} \approx 0.368$, $\frac{1}{e^2} \approx 0.135$, midpoint ≈ 0.25
- Geometric guessing: $\log_{1+\varepsilon} \Delta = O(\frac{\log \Delta}{\varepsilon})$ thresholds.
- γ-net in d dims: $\le (1 + \frac{2}{\gamma})^d$ points.
- Storing a number $\le N$: $O(\log N)$ bits; counters store $\log n$, ids store $\log m$.
- Hash k-wise over $\mathbb{F}_m$: k coefficients ⇒ $O(k \log m)$ bits.